

# **Implementation**

Cohort 3 Team 1: Pixels of Promise

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a.)

New parts of code in old files are denoted using comments, specifically:

```
// >>> NEW CODE START <<<
```

Followed by some code

```
// >> NEW CODE STOP <<<
```

Or for changed code, it is instead denoted:

```
// >>> CHANGED CODE START <<<
```

Followed by some code

```
// >> CHANGED CODE STOP <<<
```

Additionally, we have a new package called Scenarios, this includes all new files. Any new files that did not exist before start with: `// NEW: This is completely new`, or something similar.

This time, instructions to run the game are included on the GitHub's README.md document.

## **b.) Assets and Licensing**

To create the textures for our buildings, we used two free tilesets created and distributed by Game Developer Kenney ([www.kenney.nl](http://www.kenney.nl)). The packs used were his RPG Urban Pack 1.0 and his Roguelike Modern City 2.0, both of which are licensed by the Creative Commons Zero (CC0) license (<http://creativecommons.org/publicdomain/zero/1.0/>).

The CC0 license is highly suitable for our project as it places no restrictions on how we use the assets. This grants several freedoms such as the ability to modify the assets freely and customize them to fit our games style. The assets can also be used in both non-commercial and commercial projects, making it valid for our purposes.

Our project is built using libGDX, an open source game development framework licensed under the Apache License 2.0 (<https://www.apache.org/licenses/LICENSE-2.0>). This license is well suited as it is very flexible; allowing modification, distribution, and private or commercial use without restrictions. It also ensures the original authors are properly credited.

We are also using JDK 17, which is distributed under the GNU General Public License, version 2, with the Classpath Exception (GPLv2+CPE) (<https://openjdk.org/legal/gplv2+ce.html>). The Classpath Exception allows our project to link to the JDK without imposing the GPL's invasive copyleft restrictions on the rest of the project. This ensures that while we benefit from the JDK our game remains independent.

## **Unimplemented or Partially Implemented Features**

2.4 FR\_SETTINGS - The system should provide a settings menu for the user where they can alter things like the volume to change their experience of overall gameplay. We did not implement a settings menu, however the screen size changes dynamically.

2.1.8 FR\_HOVER\_INFO - Upon hovering over buildings with a cursor, information about the building and the related student satisfaction stats will be provided. There is no "hover" action, but we can click on a building or object to get more information, such as the move or sell price.

2.2.1. FR\_ALTER\_VOLUME - The user can adjust the volume of the game above and below (to the minimum, mute) the original sound setting once the game begins. We have no volume to alter as there are no sounds in the game at all.

3.2 NFR\_PLATFORM\_COMPATIBILITY -The user will be able to access the game on different types of devices, and operating systems (Linux, Mac, Windows etc.) We did not implement matrix testing to the CI but we ran the tests on Linux on GitHub Actions and most game development occurred on Apple Macs, and play tests were conducted on Windows in User Eval. So we can say that the game does work cross-platform.

3.3 NFR\_QUICK\_START - Assets should start loading before the user has decided they want to begin, so the user should not wait too long before starting the game. Our textures are not always loaded beforehand, they are sometimes lazy loaded when needed.